A.4. Factored representations

Key insight: Product states can be represented as a collection of predictions in orthogonal subspaces

There is a lossless linear projection from the submanifold of tensor-product states—represented in the joint predictive state space—to an exponentially lower number of dimensions where the beliefs over the parts are represented in orthogonal subspaces.

Although the linear span of all product states has dimension equal to the full joint space $\prod_{n=1}^{N} \dim(\boldsymbol{\mathcal{S}}_n)$, the submanifold containing all product states is of much lower dimension $\sum_{n=1}^{N} \big(\dim(\boldsymbol{\mathcal{S}}_n) - 1\big)$. There is a linear map from the joint space to a much lower-dimensional factored space. This mapping is invertible when the domain of the map is limited to the product states. See Fig. 1.

In the case of conditionally-independent factors, each *local* predictive vector $\boldsymbol{\eta}_n^{(w)}$ can be updated locally in parallel via

$$\boldsymbol{\eta}_n^{(w)} \overset{x}{\mapsto} \boldsymbol{\eta}_n^{(wx)} = \frac{\boldsymbol{\eta}_n^{(w)} T_n^{(x)}}{\boldsymbol{\eta}_n^{(w)} T_n^{(x)} \mathbf{1}_n} \,. \tag{12}$$

Rather than directly representing $\boldsymbol{\eta}^{(w)}$ in $\prod_{n=1}^{N} \dim(\boldsymbol{\mathcal{S}}_n)$ dimensions, we expect the model to directly represent

$$\boldsymbol{\eta}_{\text{FWH}}^{(w)} = (\tilde{\boldsymbol{\eta}}_1^{(w)}, \tilde{\boldsymbol{\eta}}_2^{(w)}, \ldots, \tilde{\boldsymbol{\eta}}_N^{(w)}) = \sum_{n=1}^{N} \iota_n(\boldsymbol{\eta}_n^{(w)}) \tag{13}$$

using only $d_{\text{FWH}} = \sum_{n=1}^{N} \big(\dim(\boldsymbol{\mathcal{S}}_n) - 1\big)$ dimensions, where $\iota_n : \mathbb{R}^{d_n} \to \mathbb{R}^{\sum_{n'=1}^{N}(d_{n'}-1)}$ linearly embeds each local predictive vector into the same d_{FWH}-dimensional embedding space. This is effectively just concatenating the predictive degrees of freedom $\tilde{\boldsymbol{\eta}}_n^{(w)} \in \mathbb{R}^{d_n - 1}$—which is $\boldsymbol{\eta}_n^{(w)}$ projected onto the subspace orthogonal to $\mathbf{1}_n$—for each subsystem.

In the main text, we write this less formally (but hopefully more intuitively) via the direct sum $\bigoplus$ (a notation usually reserved for vector spaces rather than vectors themselves). Context embeddings in the factored representation live in the vector space obtained from the direct sum of constituent vector spaces for each part. This corresponds to adding the vectors for each component only after they've been embedded orthogonally into this shared space.

In particular,[11] the linear embedding map takes the form $\iota_n = \sum_{m=2}^{|\mathcal{B}_n|} |b_{m,n}\rangle\langle\delta_{m-1+\sum_{n'=1}^{n-1}(d_{n'}-1)}|$, where $\langle\delta_k|$ is a row vector which is one-hot at its kth element. $\mathcal{B}_n = \{|b_{m,n}\rangle\}_{m=1}^{|\mathcal{B}_n|}$ is a set of $|\mathcal{B}_n|$ orthonormal vectors with $|b_{1,n}\rangle = |\mathbf{1}_n\rangle$. The other basis elements $\{|b_{m,n}\rangle\}_{m=2}^{|\mathcal{B}_n|}$ are rather arbitrary (besides orthogonality with $\mathbf{1}_n$) but can be obtained by Gram–Schmidt orthogonalization (or PCA on the mean-centered local predictive vectors).

In effect, the factored representation linearly embeds each local predictive vector, allotting one dimension to each degree of freedom for each predictive vector. I.e., we anticipate context-induced conditionally-independent parts represented in orthogonal subspaces. We call this the Factored World Hypothesis (FWH).

A.5. Joint-to-factored map: reduced states and relative entropy

In general, valid predictive vectors are not constrained to the product-state submanifold in the joint representation. Many different off-manifold predictive vectors would map to the same point in the factored representation—such that the joint-to-factored map is non-invertible in general. Accordingly, *while the factored representation allows for a much lower-dimensional representation, it does not naturally accommodate a representation of correlated parts* and so would correspond to a lossy representation if the parts in fact remain correlated after conditioning on the token history.

A.5.1. Marginalizing over subsystems to obtain reduced states

Given the joint predictive vector $\langle\boldsymbol{\eta}|$ and dynamics of a multipartite system, how do we take something like a "partial trace" over some subsystem(s) to obtain the reduced predictive state of the remaining subsystem(s)?

[11] It is helpful here to use bras $\langle\cdot|$ and kets $|\cdot\rangle$ to denote row vectors and column vectors, respectively. This allows for easy tracking of object type, as $|\cdot\rangle\langle\cdot|$ is a matrix and $\langle\cdot|\cdot\rangle$ is a scalar.

We achieve this via the $|\mathbf{1}_n\rangle$ vector(s) such that (i) $T_n|\mathbf{1}_n\rangle = |\mathbf{1}_n\rangle$, (ii) $|\mathbf{1}\rangle = \bigotimes_{m=1}^N |\mathbf{1}_m\rangle$, (iii) $T|\mathbf{1}\rangle = |\mathbf{1}\rangle$, and (iv) $\langle\boldsymbol{\eta}|\mathbf{1}\rangle = 1$. This allows us to "trace out" the nth subsystem via $\text{tr}_n(\langle\boldsymbol{\eta}|) = \langle\boldsymbol{\eta}|\left(\otimes_{m=1}^{n-1} I_m\right) \otimes |\mathbf{1}_n\rangle \otimes \left(\otimes_{m=n+1}^{N} I_m\right)$.

Note that this contraction is indeed functionally equivalent to the partial trace of a density matrix for quantum systems, when our systems are quantum. The procedure introduced thus generalizes the notion of partial trace to marginalize over subsystems in generalized probabilistic theories (classical, quantum, or otherwise).

This is useful for finding the closest product state to an arbitrary joint state, as measured by the relative entropy (a.k.a., Kullback–Leibler divergence):

$$\text{argmin}_{\boldsymbol{\eta}\in\boldsymbol{\mathcal{S}}_\otimes} \mathrm{D}\left[\boldsymbol{\eta}^{(x_{1:\ell})} \| \boldsymbol{\eta}\right] = \bigotimes_{n=1}^{N} \boldsymbol{\eta}_n^{(x_{1:\ell})} \,, \tag{14}$$

where $\boldsymbol{\mathcal{S}}_\otimes = \{\otimes_{n=1}^N \langle\boldsymbol{\eta}_n| : \langle\boldsymbol{\eta}_n| \in \boldsymbol{\mathcal{S}}_n\}$ is the set of product states. The answer is the tensor product of all of the reduced states, where the reduced state of the nth subsystem traces over all subsystems *except* for the nth subsystem:

$$\langle\boldsymbol{\eta}_n^{(x_{1:\ell})}| := \langle\boldsymbol{\eta}^{(x_{1:\ell})}| \left(\otimes_{m=1}^{n-1} |\mathbf{1}_m\rangle\right) \otimes I_n \otimes \left(\otimes_{m=n+1}^{N} |\mathbf{1}_m\rangle\right) . \tag{15}$$

It is important to note that the closest product state as measured by relative entropy is generally *not* the same as the closest product state as measured by Euclidean distance.

With the reduced marginals in hand, we can now quantify the **total correlation** $\mathrm{C}[\boldsymbol{\eta}]$ among all N parts via

$$\mathrm{C}[\boldsymbol{\eta}] := \mathrm{D}\left[\boldsymbol{\eta} \middle\| \bigotimes_{n=1}^{N} \boldsymbol{\eta}_n\right] , \tag{16}$$

where $\boldsymbol{\eta}_n$ is the reduced state of the nth subsystem. Eq. (16) is consistent with both classical and quantum information theory, and can be applied more generally to mixed representations.

A.5.2. Joint-to-factored map

The linear map from joint to factored representations can now be written explicitly via combining partial trace with the linear embedding maps from the previous section:

$$\langle\boldsymbol{\eta}_{\text{FWH}}^{(x_{1:\ell})}| = \langle\boldsymbol{\eta}^{(x_{1:\ell})}| \sum_{n=1}^{N} \left[\left(\otimes_{m=1}^{n-1} |\mathbf{1}_m\rangle\right) \otimes I_n \otimes \left(\otimes_{m=n+1}^{N} |\mathbf{1}_m\rangle\right)\right] \iota_n \,. \tag{17}$$

B. Minimal explicit example of joint vs. factored representations

Here, we work through the mathematics and geometry of a minimal example that compares the joint and factored representations.[12]

A nontrivial example requires at least two factors, each with at least a two-dimensional latent space. Accordingly, for the independent case, we take our minimal nontrivial example to be a process with two factors that are each non-Markovian 2-state HMMs with a binary alphabet.

We take each of these constituent HMMs to be a Simple Nonunifilar Source (SNS) (Travers & Crutchfield, 2011), with labeled transition matrices

$$T_n^{(0)} = \begin{bmatrix} 0 & 0 \\ q_n & 0 \end{bmatrix} , \quad T_n^{(1)} = \begin{bmatrix} 1-p_n & p_n \\ 0 & 1-q_n \end{bmatrix} \tag{18}$$

for $n \in \{1, 2\}$, with $p_n, q_n \in (0, 1)$. Each of these factors can be thought of as producing 'sub-tokens' from the binary alphabet $\mathcal{Z}_1 = \mathcal{Z}_2 = \{0, 1\}$. The net row-stochastic transition matrix for each factor $T_n = T_n^{(0)} + T_n^{(1)} = \begin{bmatrix} 1-p_n & p_n \\ q_n & 1-q_n \end{bmatrix}$

[12] One possibly misleading part of this minimal example is that the number of dimensions in the joint and factored representations don't seem too different. For more factors or with higher dimensions for each, the exponential discrepancy quickly becomes obvious.

has the stationary right eigenvector $\mathbf{1}_n = T_n \mathbf{1}_n = [1, 1]^\top$ (which can be thought of as integrating latent probability) and stationary left eigenvector $\boldsymbol{\pi}_n = \boldsymbol{\pi}_n T_n = \frac{1}{p_n + q_n}[q_n,\, p_n]$ (which can be thought of as the stationary distribution over hidden states if marginalizing over sub-tokens).

Observed tokens $\mathcal{X} = \{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}\}$ correspond to the possible combinations of these sub-tokens: $00 \mapsto \mathrm{A}$, $01 \mapsto \mathrm{B}$, $10 \mapsto \mathrm{C}$, and $11 \mapsto \mathrm{D}$. The minimal GHMM for this process with two independent factors has transition operators acting on the joint $\mathbb{R}^4$ space: $T^{(\mathrm{A})} = T_1^{(0)} \otimes T_2^{(0)}$, $T^{(\mathrm{B})} = T_1^{(0)} \otimes T_2^{(1)}$, $T^{(\mathrm{C})} = T_1^{(1)} \otimes T_2^{(0)}$, and $T^{(\mathrm{D})} = T_1^{(1)} \otimes T_2^{(1)}$.

The net row-stochastic transition matrix on the joint latent space $T = \sum_{x \in \{\mathrm{A,B,C,D}\}} T^{(x)} = T_1 \otimes T_2$ has the stationary right eigenvector $\mathbf{1} = T\mathbf{1} = (T_1 \otimes T_2)\mathbf{1} = \mathbf{1}_1 \otimes \mathbf{1}_2 = [1, 1, 1, 1]^\top$, (which again can be thought of as integrating latent probability) and stationary left eigenvector $\boldsymbol{\pi} = \boldsymbol{\pi} T = \boldsymbol{\pi}(T_1 \otimes T_2) = \boldsymbol{\pi}_1 \otimes \boldsymbol{\pi}_2 = \frac{1}{(p_1+q_1)(p_2+q_2)}[q_1 q_2,\, q_1 p_2,\, p_1 q_2,\, p_1 p_2]$ (which can be thought of as the stationary distribution over joint hidden states if marginalizing over token observations). We take the initial predictive vector to be the stationary state $\boldsymbol{\eta}^{(\varnothing)} = \boldsymbol{\pi} = \boldsymbol{\pi}_1 \otimes \boldsymbol{\pi}_2 = \boldsymbol{\eta}_1^{(\varnothing)} \otimes \boldsymbol{\eta}_2^{(\varnothing)}$.

We can now compare two candidate representations:

(i) The **joint representation**, where each predictive state $\boldsymbol{\eta}^{(x_{1:\ell})} = \frac{\boldsymbol{\eta}^{(\varnothing)} T^{(x_{1:\ell})}}{\boldsymbol{\eta}^{(\varnothing)} T^{(x_{1:\ell})} \mathbf{1}}$ lives on the curved 2d manifold of product states $\big\{[p, 1-p] \otimes [p', 1-p'] : p, p' \in [0, 1]\big\}$ within the 3-dimensional 3-simplex[13] Δ^3 embedded in the joint space $\mathbb{R}^4$. The collection of context-induced predictive states $\{\boldsymbol{\eta}^{(w)} : w \in \mathcal{X}^*\}$ *spans* $\mathbb{R}^4$ but only *varies* in three directions since differences in predictive states are all orthogonal to $\mathbf{1}$—since $(\boldsymbol{\eta}^{(w)} - \boldsymbol{\eta}^{(w')})\mathbf{1} = \boldsymbol{\eta}^{(w)}\mathbf{1} - \boldsymbol{\eta}^{(w')}\mathbf{1} = 1 - 1 = 0$.

(ii) The **factored representation**, where a linear transformation of the predictive state preserves all information (via a one-to-one mapping, *as long as the predictive state lives on the product-state manifold*) while producing a lower-dimensional representation. The factored representation allocates an orthogonal subspace to the latent probability simplex of each factor. In this case, since each factor has a two-dimensional latent space, the factored representation accommodates each 1-simplex with its own dimension. Since $[1, -1]^\top$ is orthogonal to $\mathbf{1}_n = [1, 1]^\top$, the degree of freedom for each factor is the scalar coefficient for $|b_{2,n}\rangle = [1, -1]^\top$. The factored representation takes the form

$$\boldsymbol{\eta}_{\mathrm{FWH}}^{(w)} = \iota_1(\boldsymbol{\eta}_1^{(w)}) + \iota_2(\boldsymbol{\eta}_2^{(w)}) \tag{19}$$

where $\iota_n : \underbrace{\mathbb{R}^{d_n}}_{\mathbb{R}^2} \to \underbrace{\mathbb{R}^{\sum_{n'=1}^{N}(d_{n'}-1)}}_{\mathbb{R}^2}$ linearly embeds each local predictive vector into a shared 2-dimensional embedding space. In particular, the embedding map takes the form $\iota_n = |b_{2,n}\rangle\langle\delta_{1+n}| = \frac{1}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix}\langle\delta_{1+n}|$, where $\langle\delta_k|$ is a row vector which is one-hot at its kth element. Altogether, this yields the factored representation

$$\boldsymbol{\eta}_{\mathrm{FWH}}^{(w)} = \tfrac{1}{2}\boldsymbol{\eta}_1^{(w)} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} + \tfrac{1}{2}\boldsymbol{\eta}_2^{(w)} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} . \tag{20}$$

Note that all points in the factored representation live in the square (ranging from $-\frac{1}{2}$ to $\frac{1}{2}$ on each side), which is just a global shift of the Cartesian product of the constituent 1-simplices $\Delta^1 \times \Delta^1$.

Fig. 5 shows these two candidate ground-truth representations side by side.

B.1. Dependent chain

A minimal dependent case with nontrivial candidate geometries can be obtained via a simple adaptation of the previous independent example. The first factor remains the 2-state SNS HMM. However, the second factor now depends on the first.

One way to realize this is by making the second factor an *input-dependent* HMM with conditioned transition matrices $T_2^{(z^{(2)}|z^{(1)})}$, where the transition parameters p_2 and q_2 now depend on the output $z^{(1)}$ of the first factor. Observed tokens $\mathcal{X} = \{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}\}$ again correspond to the possible combinations of these sub-tokens: $00 \mapsto \mathrm{A}$, $01 \mapsto \mathrm{B}$, $10 \mapsto \mathrm{C}$, and $11 \mapsto \mathrm{D}$. However, the minimal GHMM for this process with one independent and one dependent factor has transition

[13] Recall that the n-simplex, $\Delta^n = \{(v_j)_{j=1}^{n+1} \in \mathbb{R}^{n+1} : \sum_{j=1}^{n+1} v_j = 1 \text{ and } v_j \geq 0\}$, corresponds to the set of all probability distributions over $n+1$ elements. Above, the 3-simplex is the tetrahedron containing all probability distributions over the four joint hidden states of the minimal GHMM. Similarly, the 1-simplex is a line segment serving to contain all probability distributions over the two hidden states of one of the SNS factors, if we marginalize over the other factor.